Queues under Stochastic Priority Switching

Model Analysis, Numerical Replication, and Extensions

Stochastic Operations Research — Course Project

June 2026

Palmer, G. I., Panayidis, M., Knight, V., & Williams, E. (2026). JORS.

Motivation: When FIFO Fails

Real-world problem

- Surgical endoscopy waiting lists
- Patients' urgency **changes** while waiting
- Observed service order $\neq$ arrival order
- Large "overtakes": bumped up & bumped down

Research question

Can **stochastic priority switching** explain observed non-FIFO behavior when the true scheduling rule is unknown?

Core idea: Model an unknown service discipline as a random walk over priority classes.

Queueing Model

M/M/c with K priority classes

- λ_k: arrival rate of class k
- μ_k: service rate of class k
- c: number of servers
- Lower index = higher priority
- Preemptive priority

Switching matrix $\Theta = (h_{ij})$

- $h_{i,i-1}$: upgrade rate
- $h_{i,i+1}$: downgrade rate
- h_{ij} $(|i-j| > 1)$: skip-grade jumps
- Exponential switching times
- $\Rightarrow$ memoryless CTMC

Customers randomly switch priority classes while waiting — both **up** and **down**.

Methods: Dual Approach

1. Discrete-Event Simulation (Ciw)

- Poisson arrivals + Exp service + Exp switching
- Preemptive priority events
- Flexible: supports non-exponential distributions
- Noisy: requires many trials

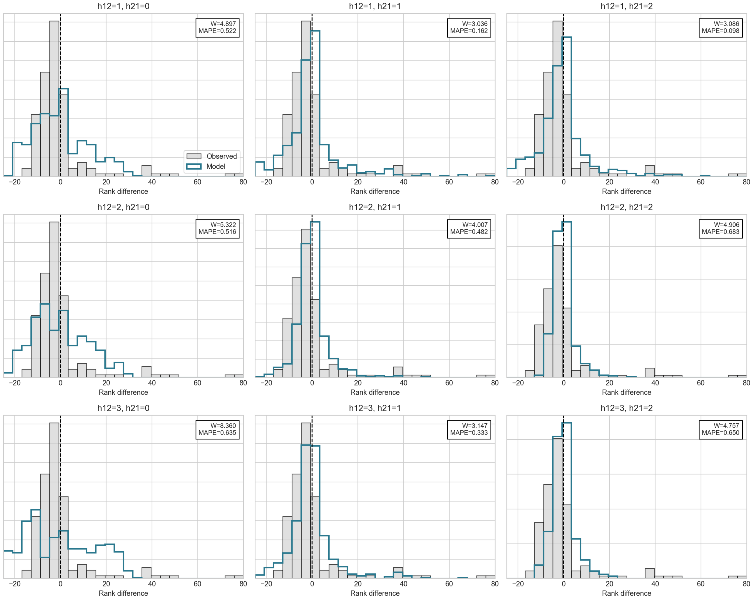
2. Dual Markov Chain (analytical)

- **Ergodic chain:** state $s = (s_0, \dots, s_{K-1})$
- $\rightarrow$ steady-state L_k, queue composition
- **Absorbing chain:** tagged customer
- $\rightarrow$ sojourn time W (phase-type distribution)

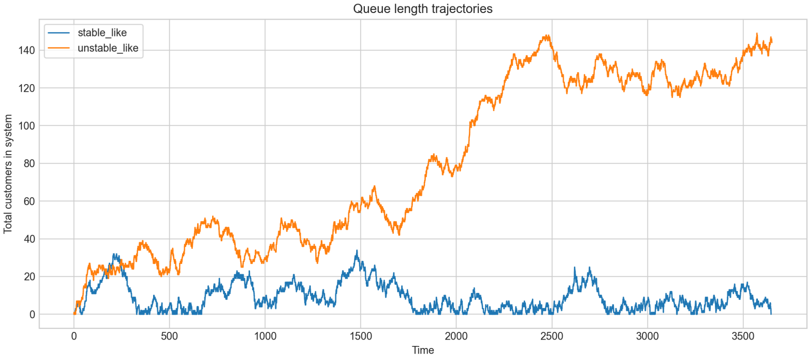
Challenge: infinite state space

Truncate with **finite bound** b $\rightarrow$ validate via $Q(b)$ and $P(b)$.

Reproduction 1: Overtaking Distribution (Fig. 5)



Reproduction 2: Stability Analysis

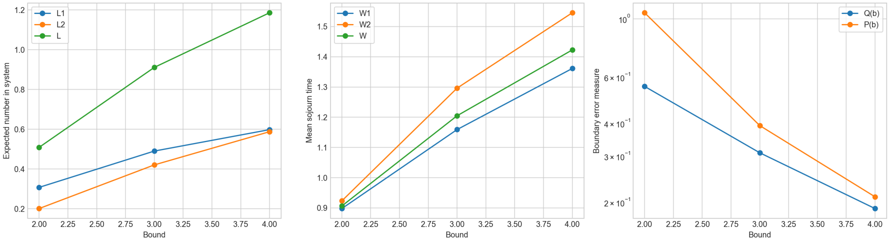


Stable-like: $\mu_1 = \mu_2 = 0.5$
ADF $p < 0.05$, queue remains bounded

Unstable-like: $\mu_1 = 0.4, \mu_2 = 0.6$
ADF $p \gg 0.05$, queue grows indefinitely

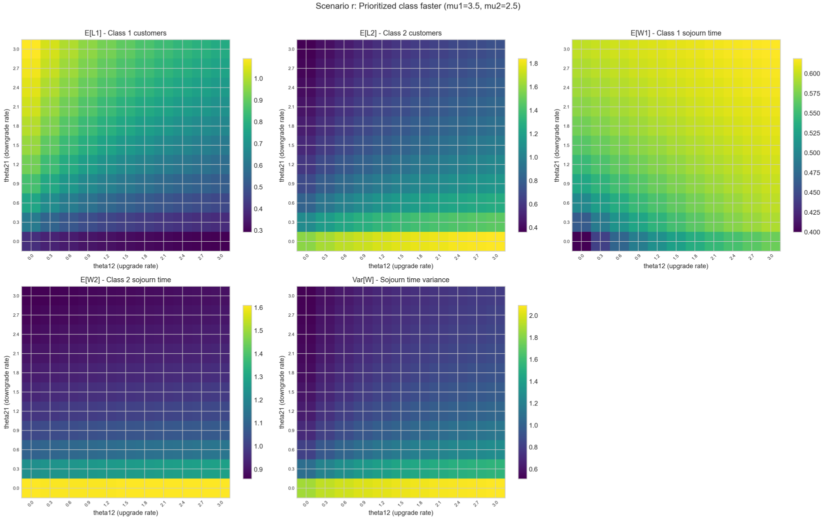
Confirms Proposition 1: stability gap region exists where Θ matters.

Reproduction 3: Bounded Markov Approximation



As $b \uparrow$: L, W converge | $Q(b)$, $P(b)$ decay exponentially (log scale).

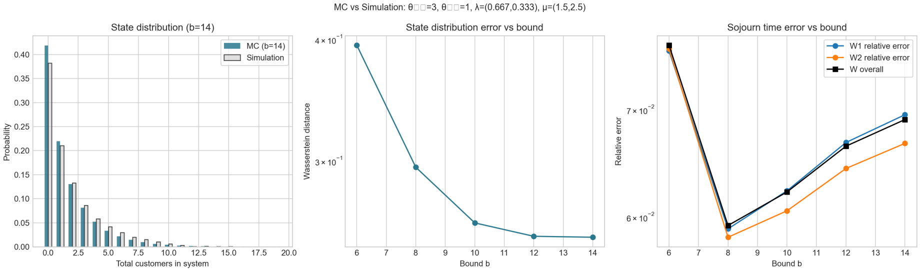
Reproduction 4: Effect of Θ Matrix



Scenario C: prioritized class has **faster** service ($\mu_1 = 3.5, \mu_2 = 2.5$)

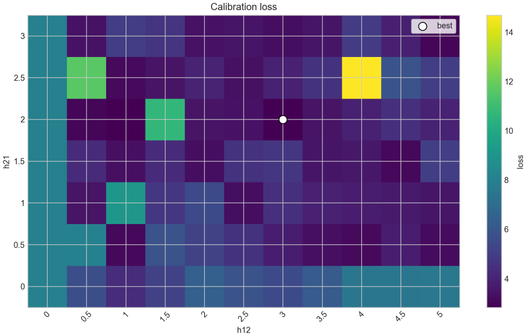
L_k, W_k, $\mathrm{Var}[W]$ heatmaps over $\theta_{12} \times \theta_{21}$ grid

Validation: Markov Chain vs Simulation



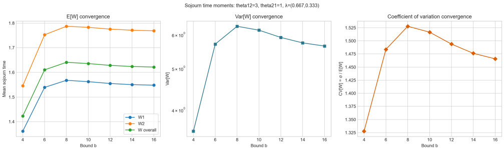
State distribution matches (Wasserstein) | Sojourn time error $\to 0$ as b grows | MC and DES converge.

Extension 1: Fine-Grid Θ Calibration

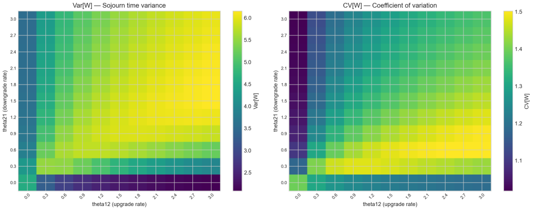


Paper: coarse 3×3 grid $\rightarrow$ **Our work**: dense 11×7 grid
Loss surface is smooth with a unique, well-defined minimum
$\Rightarrow$ overtaking-based fitting is well-behaved and identifiable

Extension 2: Sojourn Time Variance



Var$[W]$ and CV$[W]$ converge with b



Var$[W]$ and CV$[W]$ over Θ space

Phase-type absorbing chain gives **second moments** — beyond the paper's mean-only analysis.

Summary & Discussion

Reproduced (4 experiments)

- Overtaking distribution fitting (Fig. 5)
- Stability / ADF test (Proposition 1)
- Bounded Markov convergence ($Q(b)$, $P(b)$)
- Effect of Θ matrix (3 scenarios)

Extensions beyond the paper (3 items)

- Systematic MC vs Simulation validation
- Fine-grid calibration of Θ (loss landscape)
- Sojourn time variance via phase-type moments

Limitations: Exponential assumptions | CTMC state-space explosion | ADF is empirical

Future work: Non-exponential switching (Erlang, Weibull) | Multi-server | Real data fitting

Thank you!

Original paper: Palmer et al. (2026), *JORS*

Code: `github.com/geraintpalmer/DynamicClasses`

Reproduction: `github.com/[repo]`

Questions & Discussion